

Preface

Who is This Book For?

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students ¹. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people ².

Introduction courses to fundamental mathematical topics usually look like this: here're some basic definitions, let's derive some specific results from them - now go apply these in your field.

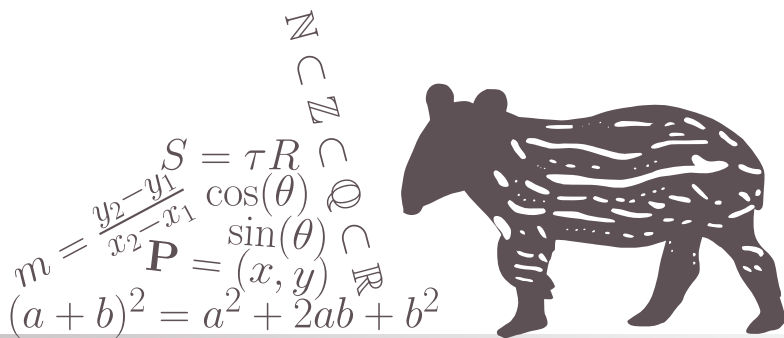
¹and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

²the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

THE BASICS OF LINEAR ALGEBRA



1.1 VECTORS AND VECTOR SPACES

1.1.1 Motivation

Motivating Problem 1.1 Gravity Force

Consider a simple simulation of classical (“Newtonian”) gravity, say of a planet with mass m orbiting a star with mass M (where $M \gg m$). At any given moment, the force acting on the planet due to the star’s gravity has magnitude

$$F = G \frac{Mm}{r^2},$$

where r is the distance between the planet and the star at that moment, and G is a universal constant (specifically, $G = 6.674 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$).

However, since we live in 3-dimensional space^a, we also want this force to be pointing in the direction from the planet to the star. To do this, we need to understand how to describe orientations in 3-dimensional space, and be able to pack together both orientation and magnitude in the correct way.

^aallegedly.

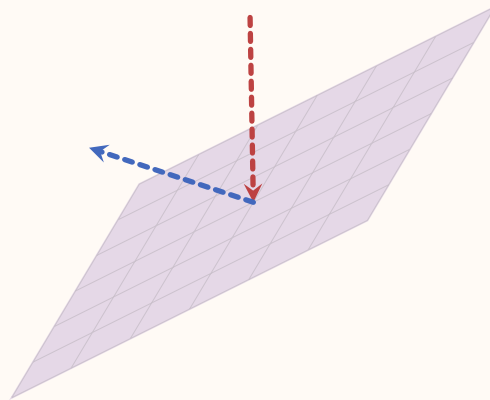


Motivating Problem 1.2 Bouncing/Reflecting Off a Wall

Imagine we are writing a 3D computer game in which some object is bouncing off a wall. Alternatively, consider 3D graphics in general, where one simulates rays of light reflecting off of smooth surfaces. In both cases the same problem needs to be solved: given a flat plane of some orientation in space, and an object hitting it from some direction - what would be the direction of the object after it bounces off the plane?

continues in the next page ➡

 continues from the previous page



To solve this problem we will need to know how to represent quantities like position, velocity and orientation in 3D, and more generally points, lines and planes - and to know how to calculate the intersection between these different objects. By the end of this section you should be able to solve this problem by yourself, and even generalize it to any number of dimensions.



Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

1.1.2 Geometric Properties of Vectors

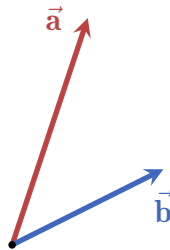
Text text text...(what are vectors, pictorial representation, notation, all starting from the origin, etc.)

Vectors have two important properties: they can be added together and be scaled

CHAPTER 1. THE BASICS OF LINEAR ALGEBRA

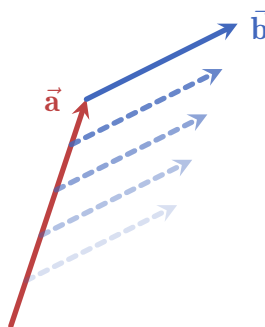
to form other vectors.

- **Adding vectors together:** any two vectors of the same dimensionality (e.g. 2-dimensional, 3-dimensional, 4-dimensional, etc.) can be added together, and the result is a third vector. Pictorially, here are two vectors $\vec{a}$ and $\vec{b}$ which we want to sum together:

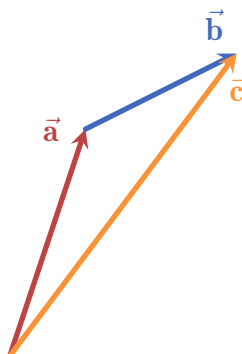


The steps to produce the sum of $\vec{a}$ and $\vec{b}$ is the following:

1. move $\vec{b}$ such that its origin aligns with the head of $\vec{a}$:

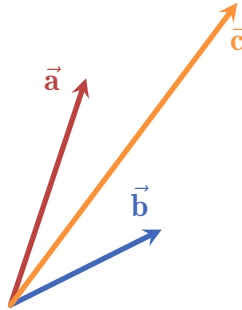


2. draw a new vector from the origin to the head of the shifted $\vec{b}$:

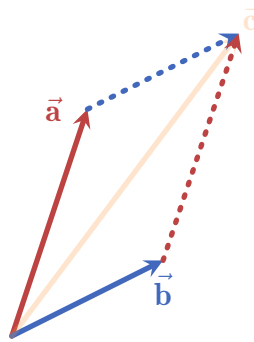


1.1. VECTORS AND VECTOR SPACES

3. the resulting vector $\vec{c}$ is the sum of $\vec{a}$ and $\vec{b}$. Here are all three vectors together from the same origin:

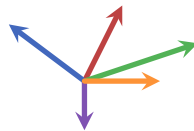


This method is called the **triangle method**, and it displays an important property of vector addition: it is commutative! This means that the order of addition doesn't matter: $\vec{a} + \vec{b} = \vec{b} + \vec{a}$. To see this visually, we can apply the triangle method twice, i.e. moving both $\vec{a}$ and $\vec{b}$:



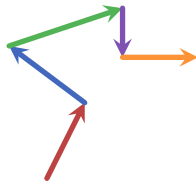
We see that no matter which vector we add onto which (i.e. what is the order of addition) - the result is the same. The shape formed by this demonstration is called a **parallelogram**, and it is incredibly important in linear algebra as a whole as we will see in future sections.

This method of summing vectors can be applied to any number of vectors. For example, consider the following 5 vectors:

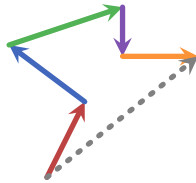


To sum them together, all we need is to start with one of the vectors (say, the red one) and use the triangle method to add the other vectors step by step:

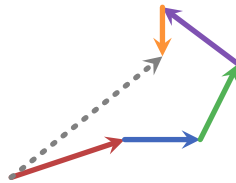
CHAPTER 1. THE BASICS OF LINEAR ALGEBRA



Finally, we connect the origin of the first vector with the head of the last vector (in this case the orange one):



and that's it, the gray dashed vector is the sum of all five vectors. To see that the order of addition really doesn't matter we can do this again but this time starting with the green vector and changing the summation order completely:



1.1.3 Common Coordinate Systems in 2- and 3-Dimensions

1.1.4 Geomteric Objects

1.1.5 Projections and Rejections

1.1.6 Vector Spaces

1.1.7 Solving the Motivating Problems

1.1. VECTORS AND VECTOR SPACES

Actual page layout values.

<code>\paperheight = 845.04684pt</code>	<code>\paperwidth = 597.50787pt</code>
<code>\hoffset = 0.0pt</code>	<code>\voffset = 0.0pt</code>
<code>\evensidemargin = 35.28255pt</code>	<code>\oddsidemargin = -0.56833pt</code>
<code>\topmargin = -10.73834pt</code>	<code>\headheight = 18.125pt</code>
<code>\headsep = 21.75pt</code>	<code>\textheight = 591.5302pt</code>
<code>\textwidth = 418.25368pt</code>	<code>\footskip = 50.75pt</code>
<code>\marginparsep = 12.8401pt</code>	<code>\marginparpush = 6.52495pt</code>
<code>\columnsep = 10.0pt</code>	<code>\columnseprule = 0.0pt</code>
<code>lem = 12.0pt</code>	<code>lex = 5.172pt</code>